BASE CHANGE FROM A SEPARABLY ALGEBRAICALLY CLOSED FIELD INDUCES A BIJECTION ON CONNECTED COMPONENTS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-varieties/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup, of the native theorem declarations with mathematical macros, and of the project's own no-op definitions of the reference display environments; no macro was invented or redefined.

Editorial note. The two same-chapter references are located at their \label lines inside the pinned chapter file; the cross-chapter pointer is located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim line carrying its canonical tag.

Allowed standard prerequisites (editorial).

- Over a separably algebraically closed field, a field extension is geometrically connected. Algebra, Lemma labelled algebra-lemma-separably-closed-connected-implies-geometric (canonical Stacks tag 037U); invoked at proof lines 781-784. Supplies that k' is geometrically connected over k; the passage to the spectrum and the product decomposition are retained in the counted span.
- The spectrum of a geometrically connected algebra is geometrically connected. Same chapter, Lemma labelled lemma-affine-geometrically-connected (source line 758, Stacks tag 0386); invoked at proof lines 785-786. Supplies that $Z = \mathrm{Spec}(k')$ is geometrically connected over k, so that the component-bijection lemma applies.
- Projection from a geometrically connected factor induces a bijection on connected components. Same chapter, Lemma labelled lemma-bijection-connected-components (source line 732, Stacks tag 0385, counted separately in this batch); invoked at proof lines 787-788. Supplies the bijection on components once the factor is geometrically connected; recognizing the base change as a product with Z is the counted step.

Source reference. Original lemma 037U: sources/stacks-chow/tags.txt, source lines 4213–4213. Cross-chapter standard prerequisite invoked once at proof lines 781-784 for geometric connectedness of k' over k; the scheme-theoretic assembly on top of it is retained in the counted span.

Source reference. Original lemma 0386: sources/stacks-varieties/pinned-original-source.tex, source lines 758–758. Same-chapter pointer invoked at proof lines 785-786 to convert geometric connectedness of k' into geometric connectedness of Spec(k').

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Source reference. Original lemma 0385: sources/stacks-varieties/pinned-original-source.tex, source lines 732–732. Same-chapter pointer invoked at proof lines 787-788 to obtain the bijection on components; the identification X_{k'} = Z x_k X is retained in the counted span.

Lemma 0.1. *Let k'/k be an extension of fields. Let X be a scheme over k. Assume k separably algebraically closed. Then the morphism $X_{k'} \to X$ induces a bijection of connected components. In particular, X is geometrically connected over k if and only if X is connected.*

Proof. Since k is separably algebraically closed we see that k' is geometrically connected over k, see Algebra, Lemma 037U. Hence $Z = \mathrm{Spec}(k')$ is geometrically connected over k by Lemma 0386 above. Since $X_{k'} = Z \times_k X$ the result is a special case of Lemma 0385. □